3D Knitting

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Background

1.1 Context

This document describes the modelling approach for **3D knitting**, focusing on machines that directly knit complete 3D garments without sewing.

It ensures methodological consistency with the other knitting process models (flat, circular, hosiery) while accounting for the distinctive features of 3D knitting.

1.2 Scope and functional unit

- **Functional unit:** 1 kg of 3D knitted fabric (WHOLEGARMENT-type).
- **System boundary:** Gate-to-gate process, from yarn input to finished 3D garment.
- **Included:** Yarn losses, lubricants, electricity (direct + indirect), capital goods and their EoL, textile waste.
- **Excluded:** Fibre production, finishing, packaging, transport.

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2. Process Description

3D knitting machines (e.g., **Shima Seiki SWG-XR**, **MACH2XS**, **Stoll CMS 3D**) integrate design and garment shaping in one process, eliminating most cutting and sewing.

This results in minimal material waste but higher **specific energy demand (kWh/kg)** due to longer cycle times and lower productivity per unit mass compared to circular knitting.

Inventory Data – 3D Knitting (mix)

Category	Flow	Amount	Unit	Comment
Product	3D knitted garment	1	kg	Average of mixed yarn types (cotton/synthetic).
Technosphere flows	Dtex Yarn (“e.g. Cotton”) – yarn losses	0.02	kg	Yarn losses 2% (Laursen 2007; van der Velden 2014; Sandin 2019). Typical realisation 94–98%

Category	Flow	Amount	Unit	Comment
				(Advanced Knitting Technology, 2021; Rakib Ahmed, 2024).
	Lubricant, average	0.008	kg	**Lubricant:** 8 g kg^{-1} fabric "Sandin et al.2019 pp45 - The lubricant was assumed to correspond to 8% of the weight of the knit."
	Air emissions from lubricant	0.008	kg	Equal to lubricant mass for simplicity.
Electricity / heat	Electricity mix	4.8	kWh	Derived from machine data: 1.8 kW × 1.2 garments/h × 0.45 kg/garment → 3–5 kWh/kg (Shima Seiki SWG-XR & MACH2XS; Stoll CMS; NTNU 2024). Confirmed by manufacturer specs
Indirect energy	HVAC, compressors, lighting	4.46	kWh	Standardised value for knitting indirect systems (compressed air, cooling, robotics).
Capital goods	Building machine (market for building machine)	0.0004	kg/kg fabric	Based on 1,200–1,400 kg machine amortised over 3.6 million kg lifetime output (see below).
Waste to treatment	EoL Machine (Steel)	0.0003	kg	85% of total mass distributed by lifetime output.
	Recycling others (Other Materials)	0.0001	kg	15% of mass (electronics, polymer components).
	Textile waste incineration (no credits)	0.02	kg	Seamless production: >95% utilisation, <5% waste (K&G Garment 2025; GIZ 2023).

4. Capital Goods (Machine)

4.1 Equipment data

Machine	Weight (kg)	Source
Shima Seiki SWG-XR124 / XR154	1,300–1,400	Shima Seiki 2025 datasheet
Shima Seiki MACH2XS	1,000–1,212	Shima Seiki 2025
Stoll CMS 3D	1,200	Stoll / Karl Mayer catalogue

4.2 Allocation parameters

Scenario	Years	Hours/day	Days/year	Productivity (kg/h)	Lifetime output (kg)	Allocation (kg/kg fabric)
Average	15	16	300	5	360,000	0.0004

> Machine allocation is calculated as:
> `kg machine per kg fabric = M_machine / Q_lifetime`

4.3 End-of-Life (EoL) allocation

Material	Share (%)	Weight (kg)	Allocation (kg/kg fabric)
Steel	85	1,105	0.0003
Other materials	15	195	0.0001

4.4 ecoinvent representation

Field	Entry
Dataset	market for building machine (GLO)
Amount	2.9E-07 unit/kg
Unit	unit
Product	building machine

(Calculated from 0.0004 ÷ 1,400 kg.)

4.5 Building Infrastructure

The 3D / WHOLEGARMENT knitting process model includes an optional building infrastructure (pavilion) share as a modular and auditable input. The building is represented by an ecoinvent-style construction dataset ("building, hall") whose reference flow is m² of industrial hall.

4.5.1 Allocation principle (area amortisation over lifetime output).
Building infrastructure is attributed to 1 kg of knitted garment by amortising the allocated hall area over the lifetime production associated with the knitting cell:

$$\text{Building intensity } (\text{m}^2/\text{kg}) = \frac{A_{\text{alloc}}\ (\text{m}^2)}{Q_{\text{lifetime}}\ (\text{kg})}$$

$$Q_{\text{lifetime}} = \text{productivity (kg/h)} \times \text{hours/day} \times \text{days/year} \times \text{years}$$

where A_{alloc} is the hall area allocated to the knitting cell (machine footprint + access + yarn storage + programming workstation + maintenance space + internal logistics), and Q_{lifetime} is the total knitted output produced during the reference service life under the chosen operating regime.

Technology rationale.
3D / WHOLEGARMENT knitting produces complete, fully fashioned garments in a single operation with near-zero cut waste. However, the complexity of the knitting programme, frequent needle-bed changes, and the significantly lower knitting speed relative to circular or seamless technologies result in the lowest throughput per m² of all knitting technologies considered. This low throughput drives the highest building intensity per kg of output in the technology group.

Practical implementation.
In the 3D / WHOLEGARMENT knitting foreground unit process, we add an input: *building, hall* (unit = m²) with amount = 4.86E-04 m²/kg.

End-of-life of the building.
The building dataset includes construction, maintenance and decommissioning within the background system model. No separate building EoL module is created in the foreground; only the amortised m²/kg input is provided.

Default value (this study). In the absence of site-specific A_{alloc} and Q_{lifetime}, the following reference intensity is applied:

> 3D / WHOLEGARMENT knitting: **4.86E-04 m²/kg**

This is the highest value in the technology group, consistent with the expectation that 3D knitting has substantially lower throughput per allocated space than all other knitting technologies. Physical plausibility was cross-checked using machine envelope data from SHIMA SEIKI MACH2VS WholeGarment® (A = 3430 mm, B = 3266 mm, envelope ≈ 11.2 m²) with a cell factor of ~3×, yielding an allocated area and calculated intensity in the same order of magnitude as the adopted default.

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5. Indirect Energy

Indirect energy (HVAC, compressors, lighting, robotics) is modelled as **4.46 kWh/kg**, consistent with all knitting processes.

In 3D knitting, this category is particularly relevant due to:

- increased use of vacuum and suction for yarn handling,
- robotic arms for automated take-down and quality inspection,
- air-conditioning to stabilise yarn humidity.

The indirect-to-direct electricity ratio (~1:1) is realistic for 3D knitting environments with moderate automation.

Subsystem	Share (%)	Equivalent (kWh/kg)
Machines	35	1.56
Compressors	25	1.12
HVAC	30	1.34
Lighting	6.5	0.29
Cleaning / auxiliaries	3.5	0.16

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6. Waste and Material Efficiency

3D knitting eliminates cutting and assembly waste.

Compared to traditional garment production (10–25% cutting waste), seamless knitting reaches >95% material utilisation (GIZ, 2023; K&G Garment, 2025).

The model therefore applies **0.02 kg/kg** textile waste, equivalent to ~2% of yarn input.

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7. Key Assumptions Summary

Parameter	Average value	Comment
Electricity (machine)	4.8 kWh/kg	Based on Shima Seiki and Stoll specs Dados de Consumo Energético em ...
Indirect energy	4.46 kWh/kg	Harmonised with other knitting processes
Lubricant	0.008 kg/kg	3-8 g/kg per literature
Yarn loss	0.02 kg/kg	2% default, consistent with yarn realisation 94–98%

Parameter	Average value	Comment
Machine mass allocation	0.0004 kg/kg	Derived from 1,200–1,400 kg machine
EoL (steel)	0.0003 kg/kg	85% share
Textile waste	0.02 kg/kg	<5% material losses

Reference

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